APERIO | AUTONOMOUS EARTH INTELLIGENCE PLATFORM

# AUTHORITY DISASTER INTELLIGENCE ASSESSMENT • FLOOD

Location: Tinsukia, Assam, India • Generated: 08 Oct 2026 17:59 UTC • Status: AWAITING HUMAN REVIEW

| | | | |
|---|---|---|---|
| **Target Event** | Flood • Tinsukia | **Risk Probability** | **26% (HIGH)** |
| **Affected Area** | N/A | **Model Confidence** | 88% Indicative |
| **Exposed Population** | Assessed in loop | **Antecedent Rainfall** | 120 mm (72h: 572mm) |
| **Terrain Slope** | 15° (Steep escarpment) | **SAM Scar Boundary** | Validated (7.03 km²) |
| **Satellite Observation** | Sentinel-1 SAR + Sentinel-2 MSI | **Human Verification** | AWAITING HUMAN REVIEW |

## Multi-Agent Intelligence & Evidence Progression

**1. Satellite Ingestion:** Copernicus Sentinel-1 C-SAR IW mode acquired observation passes over Tinsukia. Radar backscatter penetration enabled continuous surface change mapping.
**2. DEM Terrain Modeling:** Horn algorithm computed local terrain slope of 15° across drainage lines.
**3. SAM Spatial Segmentation:** Delineated spatial impact boundary encompassing high-risk terrain.
**4. XGBoost & Bayesian Sequential Update:** Domain evidence likelihoods updated sequential prior belief to yield calibrated posterior risk of **26%**.

## Response Recommendations & Relocation Planning

| Designation | Capacity / Occupancy | Distance / Road Status | Hazard Exposure |
|---|---|---|---|
| Designated Relief Center (Tinsukia) | 650 / 380 (270 Available) | 2.4 km • Open | LOW RISK |
| Community Relief Hall West | 420 / 310 (110 Available) | 4.1 km • Standby | MODERATE |
| District Camp North | 800 / 220 (580 Available) | 6.8 km • Clear | LOW RISK |

## Accountable Operator Review & Audit Trail

**Reviewing Desk:** APERIO Response Operations Desk
**Operator Notes:** Automated verification report
**Operational Boundary:** Decision Support Output only. All evacuations require local ground confirmation.